

Claude:

Asked Claude to generate a basic structure and help with the approach, what 3d files to use, etc.

Used Claude for help with misc. Tasks such as fixing small bugs, helping with the readme files and data flow, creation of the data, and the final prompt reports.

VS Code (Github Copilot):

Used to redesign the approach, and test various styles and approaches before settling on a single approach.

Prompt 1:

I'm building "Kohler AI Bathroom Designer" — a hackathon project (solo, tight deadline) that recommends bathroom fixture bundles based on a customer's room dimensions, budget, and aesthetic theme, sourced from a real Kohler product catalogue (466 products across toilet, mirror, washbasin, faucet, shower, bathtub categories).

Current stack:

- Backend: FastAPI + SQLAlchemy + SQLite (auto-seeds from a CSV catalogue on startup)
- Frontend: React (Vite)
- AI: Gemini API for two tasks — (1) matching a coherent product bundle to a budget/theme, and (2) enriching the catalogue with style tags and estimated footprint dimensions (`width_in`, `depth_in`) per product, since the source price book doesn't include those

Two existing user flows:

1. **Generate a design (top-down):** user enters room width/depth (ft), budget (INR), and picks an aesthetic theme (Minimalist Modern / Classic Luxury / Japanese Zen) via `DesignForm.jsx`. This calls `POST /design/`, which filters the catalogue by budget (`catalog_filter.py`), asks Gemini to pick one product per category that's budget-fitting and style-coherent (`ai_matcher.py`), and returns a bundle + a room layout. The layout is currently generated as a flat SVG string by `layout_generator.py` (simple rectangles placed left-to-right, not spatially accurate).
2. **Browse and match (bottom-up, manual selection):** user browses the catalogue by category (`CatalogBrowser.jsx`, backed by `GET /products/`), manually selects one product they like, and the app recommends complementary items from every *other* category that share style tags or collection with the selected item (`GET /products/{sku_code}/similar`, scored in `similarity.py` by tag overlap). Results show in `SimilarItems.jsx`.

The task:

Replace the current flat SVG layout with a real interactive 3D visualization using **React Three Fiber** (`@react-three/fiber`) and **drei** (`@react-three/drei`) on top of Three.js.

Requirements:

- Render the room as a floor plane sized to the user's actual `room_width_ft × room_depth_ft` input.
- Render each selected/recommended product (from *either* flow — the generated bundle, or the manually-selected item plus its complementary matches) as a simple primitive shape (box or

cylinder is fine — no need for realistic fixture models), sized proportionally using that product's width_in/depth_in from the catalogue data, and positioned so items don't overlap.

- Color-code shapes by category (e.g. toilet, washbasin, faucet, shower, bathtub each get a distinct color) and add a floating text label above each one showing the product's model_name, using drei's <Text> component.
- Add OrbitControls (from drei) so the user can rotate/zoom/pan the 3D scene.
- The backend's layout_generator.py needs to change from returning an SVG string to returning structured JSON instead — an array of objects like {sku_code, category, model_name, x, y, width_in, depth_in} — so the frontend can actually place objects in 3D space rather than just embedding a flat image.
- This should work for both existing flows: the generated bundle (already has all selected items together) and the manual browse-and-match flow (anchor item + its complementary picks across categories).

Prompt 2:

ok so now the 3d visualization isnt working, and i want it to be interactive such that you can move around in it, in a specific window, and it should span the entire width of the site, like a top orange bar, and you enter your dimensions below that, and a visualization is generated where when you scroll down, you can interact with it, such as camera movement, and moving in a certain direction, also maybe on each object, the white dot marker should come that's subtle, like in video games, and when you click on it, you can see the specifications and a menu to modify it

Prompt 3:

ok, the new updated file is catalogue.csv, with the object files in the folder named 3d_files. For now there are 4 categories - toilet, toilet_seat, wash basin and faucet,. Update the structure everywhere for that as the new data source. Ignore the bath for now, but display the 3d models of the object as the image in the custom selection menu. In the 3d view, put all of the objects in a place, in a traditional bathroom design. The sku_code and file names match, however there are no object files for faucets, we won't be displaying those. For the sinks, make a white box at the bottom. The vessel sinks sit on top, and keep about a 1 inch gap on each side of the sink. So essentially a platform. The toilet seats should line up approximately with the toilet. Keep some space for the bathtub area. the dimensions and price are provided along with the style tags, Make sure its consistent across all files with this new system, not from catalogue_seed

Antigravity IDE (Gemini 3.6 Flash):

Used Antigravity for the majority of the project.

Essentially redesigned the entire 2d and 3d layout editors, as well as the base UI to keep it more minimalistic. Helped with orientation of the 3d objects and the intersection objects. Made specific rulesets for the object placements (eg. the toilet needs to snap to a wall, etc.). Updated the product bundle recommendation system.

Prompt 1: Project context:

I'm building "Kohler AI Bathroom Designer" — a hackathon project (solo, tight deadline) that recommends bathroom fixture bundles and shows a 3D visualization of the resulting layout, sourced from a curated, hand-picked Kohler product dataset.

Current stack:

Backend: FastAPI + SQLAlchemy + SQLite (auto-seeds from a CSV on startup) Frontend: React (Vite) + React Three Fiber (@react-three/fiber) + drei (@react-three/drei) for the 3D scene AI: Gemini API for bundle matching and catalogue enrichment

Data source change — replace catalog_seed.csv everywhere with catalogue.csv:

The catalogue has moved from a large, auto-extracted 466-product CSV to a small, hand-curated one. Update every reference across the codebase (seed.py, the Product model, any hardcoded paths or filenames) from catalog_seed.csv to catalogue.csv.

New CSV schema (catalogue.csv):

sku_code, category, model_name, style_tags, description, price_inr, height_in, width_in, depth_in

style_tags is semicolon-separated (e.g. minimalist;modern). Dimensions and price are provided directly in this file — don't estimate or enrich them further.

Categories — only four now, not six: toilet, toilet_seat, washbasin, faucet. Drop mirror, shower, and bathtub from all category lists, filters, and dropdowns across the codebase (frontend category selector, backend filtering logic, everywhere a category enum/list currently exists) — bathtub specifically is deferred, not removed conceptually (see layout note below), but the others are fully out of scope for now.

3D model files — new folder: 3d_files:

OBJ files live in 3d_files/, and filename == sku_code + ".obj" — exact match, same convention as before, just a renamed/relocated folder. Update any path references accordingly (e.g. if previously frontend/public/models/, move to frontend/public/3d_files/ or wherever this folder actually lives in the repo — confirm the real path and update consistently).

Important exception — faucets have no OBJ files: don't attempt to load or display a 3D model for any faucet, and don't show a broken/missing-model state for them either — just exclude faucets from 3D rendering entirely. They still exist in the catalogue/pricing/matching logic, just not in the visual scene.

Catalogue browser — use the 3D model as the thumbnail image:

In the manual/custom selection menu (product browser), instead of a text-only list or a placeholder icon, render a small live 3D preview (or a pre-rendered snapshot) of each product's OBJ file as its thumbnail. For products with no OBJ (faucets), fall back to a plain placeholder or text card — don't break the grid.

3D scene layout — arrange as a traditional bathroom, not a random/grid placement:

Specific placement rules:

Wash basin: render as a vessel sink sitting on top of a plain white box platform/counter. The platform should be sized so there's roughly a 1-inch gap between the sink's footprint and each edge of the platform (i.e. platform width/depth = sink width/depth + 2 inches, centered under it). Toilet seat: must be positioned so it lines up (approximately) with the toilet's position — i.e. toilet_seat and toilet share roughly the same x/z placement, seat sitting logically on/near the toilet rather than placed independently elsewhere in the room. Bathtub area: even though bathtub products aren't in the catalogue yet, reserve a clear zone in the room layout for a future bathtub placement — don't fill that space with other fixtures, so it can be dropped in later without relayout. Overall arrangement should read as a believable room layout (fixtures against walls where sensible, not overlapping, roughly matching how a real bathroom is organized) rather than a neutral evenly-spaced grid.

Consistency requirement: apply this new catalogue.csv schema and the four-category structure uniformly — backend model, seed logic, filtering, frontend category lists, catalogue browser, and the 3D scene should all agree on the same categories and the same data source. Nothing should still reference the old catalog_seed.csv six-category structure.

Bathroom Planner UI Redesign — Prompts

Prompt 1: nvm remove the plant, also i want the interaction menus to be transparent like before, just with cleaner and sleeker text. for the bathtub, the menu seems to go out of bounds, so maybe keep the menu coming from under the circle, or adjust the angle. The base position of the mirror also intersects with the window.

Prompt 2: remove the plant 3d model btw

Prompt 3: Also make a menu to add basic elements such as a mirror, window, towel bar, dustbin, etc.

Prompt 4: remove the emojis, and put the floor surface and wall surface below the selected tool, at the bottom, so everything fits

Prompt 5: I have updated the catalogue with the new categories, as well as the colour/finishes. Update the required files so that the actual colours are visible. For faucets, there are 2 categories - base-mount and wall-mount, position the faucets centrally from the sink accordingly, either at the base or on the wall. For the bathtubs, 2 categories -> free standing and drop in. For free standing, keep them on the floor, and for drop in, make a platform and make the bathtubs be inside that, like irl. The border should be about 1 inch on all sides of the tub for the drop in tubs. By default, you can include a bath section without the bathtub, add an option for the user to select either, and if the bath section is selected, 2 glass walls like a corner bath setup, or if it spans the entire width of the room, one glass wall. Make sure it feels like an actual bathroom. The glass wall should have a door of course. For the towel arm, there are a few options. Adjust the height at which they're placed according to real heights. Make sure all the products are aligned and realistically placed when added.

Prompt 6: when the bathtub zone is selected, make a bathtub area, and fit one if possible. Also the product catalogue is empty. I had put a new file for catalogue.csv, make sure it's updated everywhere.

Prompt 7: when i open the product catalog, everything goes white, and the models stop showing

Prompt 8: instead of the colours, showcase the item itself. Also the faucet object files havent loaded in the catalog browser. Make sure the faucets dont intersect with the sink. Align the faucet's front as well with the sink and vanity mirror. Make the interactive tags for all the element types in the object file, and align the camera so theyre properly visible. The shower is too high, so the element isnt visible. Allow the camera to properly go towards where it needs to during inspection. Also allow the free view to pan upwards, it seems to be blocked.

Prompt 9: You had added the object files before here, just do that, why did you change all of this, a fixed orientation of the object file in the finish that it was in earlier, just do that

Prompt 10: remove the category from the image in the catalogue, also the camera goes somewhere else in the 3d view, fix it to get it back to how it was before, it when i click on the sink and then the faucet, it should directly go from the sink angle to the faucet angle, and then return smoothly backwards, like before.

Prompt 11: The items that are already added, instead of add to layour the button should say already added and should be a diff colour, like blue. Also some of the images arent fitting in

Prompt 12: When i click on the shower area, the screen becomes blank and i cant see anything

Prompt 13: When clicking on to and off the inspection, the camera should be smooth like earlier todsay. Right now it doesnt follow the exact same path, and faces the wrong direction sometimes. It is smoother than before but it should work like an interactive game where you click on an item, and it takes you to that item, and click off, and goes backwards back to where you were earlier. Also, when clicking on the shower inspection, it is still out of the fov, and the camera is too low for that.

Prompt 14: isInspecting is not defined

Prompt 15: Again the shower section isnt working properly, it isnt showing on the screen. It is always above the camera

Prompt 16: The issue is the centered hotspot is too low for the shower section, or the showerhead. Siunce the base height itself is higher than other componjents.

Prompt 17: the dot for the sink should be closer to the sink slightly but the shwoer issue still isnt fixed. Because the entire bath region has a camera focus point i guess. Focus it on the shower head itself, if theres no bathtub

Prompt 18: still zooms in here

Prompt 19: in the 2d planner, the default 0 in the budget doesnt move or get erased. Instead keep the budget blank. Also when switching in the 2d plannner between the bathtub and other components, it should be updated by default, and other elements should move to an empty space or be deleted. If a component clearly doesnt fit, it should show a message that it doesnt fit. Make sure it dynamically arranges everything. Also have a reload generation button next to reset layout. on the top bar

Prompt 20: for the bathtub zone, dont have the shower glass, just the actual bathtub if it fits int hebudget. The reload generation should not reset the 2d layout but reload the 3d layout w.r.t the new changes made in the 2d layout and the changed bundle

Prompt 21: without affecting the camera for any of the other aspects, make the entire bathroom section visible in the camera, zoom such that if changes need to be made in any component it is possible

Prompt 22: the bathrub zone is vertical but the bathtub is horizontal and interferes with the toilet. Also the already added section only wworks for toilets, not the other components. Make Keep the camera

movement the same but examine the 3d models properly and make extra platforms if required to ensure that everything lines up like an actual bathroom designer. You can make changes to how the 2d or 3d stuff is defined, but make sure the core functionality and camera smoothness remains the same

Prompt 24: don't do added by name, do it by sku, if 2 have the same name it doesn't work. Also the bathtubs don't switch and the orange arrow should overlap the interaction points. The toilet is inside the thing, and the toilet should also snap to a wall. And not be allowed to have the back facing elsewhere.

Prompt 25: in the 2d planner, the shower zone should also show up, and they should not be allowed to intersect with the toilet, even when switching between the 2

Prompt 26: the toilet is still oriented wrong, and the 2nd image shows what happens when i right click on the shower section, its dimensions change and take over the entire left side including the toilet

Prompt 27: the washbasin and faucet aren't showing up, and the bathtub shows already added when the shower enclosure is active, in the 2d catalogue. The faucet zoom also is wrong, reverse that back to how it was before these changes

Prompt 28: for the drop in category, the bathtubs should be inside the platform, also make sure that if it doesn't fit it displays a message, or moves the components it intersects with, whatever is more feasible. Do not allow a changed bathtub to intersect with existing elements.

Prompt 29: the mirror should always be behind the sink, even like this pic, but should never mix in with the window

Prompt 30: also the faucet should always come from the mirror side of the sink, not sideways

Prompt 31: this one has the front facing up. Also make the gold colour more light, like a yellowish gold, same with bronze. Also add a platform for the faucet to rest on behind the sink

Prompt 32: the faucet orientations of the other ones are wrong, make the base mounted ones be base mounted in the sub category, and wall mounted be on the wall

Prompt 33: ok now based on this stuff, revamp the allocation, when you input the room requirements, also add a generate button to allocate and design the room according to those specs such as room length, width, budget, requirements for the setup and aesthetic theme. Make sure you change the floor and wall surface theme too with that. Make a proper allocator system.

Prompt 34: don't put emojis, and keep the same themes that are there currently, just add a button below for the generation, and leave the rest as is. I meant more so change the allocation engine to be as good as possible w.r.t the requirements, and make it unique, based on the dimensions of the products, description, tags and colour schemes. Also alter the floor and wall surface themes when allocating. Make sure it all goes well together in terms of design, colours, etc.

Prompt 35: ok now based on this stuff, revamp the allocation, when you input the room requirements, also add a generate button to allocate and design the room according to those specs such as room length, width, budget, requirements for the setup and aesthetic theme. Make sure you change the floor and wall surface theme too with that. Make a proper allocator system. don't put emojis, and keep the same themes that are there currently, just add a button below for the generation, and leave the rest as is. I meant more so change the allocation engine to be as good as possible w.r.t the requirements, and make it unique, based on the dimensions of the products, description, tags and colour schemes. Also alter the floor and wall surface themes when allocating. Make sure it all goes well together in terms of design, colours, etc.

Prompt 36: look, in the existing UI, we just need to add a generate button below the room dim and specs box on the left side. It should be in line with the rest of the ui, no emojis. All it needs to do is give a product bundle and update the selected products summary according to the selected fields. and design the 3d bathroom in such a way. It should change the floor and wall themes, etc. based on the aesthetic theme given. The product skus have tags. So essentially, just update and refine the allocation system, and add the generate button. And no, dont hardcode the bathtub, detect it from the mode.

Prompt 37: make sure that the faucets clear the height of the veil tall basin